

A tensor conductivity law that removes the first-order diffuse-interface bias from homogenized $\mathbf{K}_{\text{eff}}$

Abstract

We compute the effective thermal conductivity $\mathbf{K}_{\text{eff}}$ of evolving snow microstructures by solving the steady-state homogenization cell problem of Calonne et al. (2011) directly on the phase field, where the ice–air interface is a diffuse band a few ε wide rather than a sharp surface. Inside that band the conductivity must be interpolated, and the standard arithmetic mixture law lets the band’s sub-layers conduct in parallel in every direction, although across the interface they are stacked in series; the band therefore short-circuits the interface’s thermal resistance, with an error proportional to ε . At $K_{\text{i}}/K_{\text{a}} \approx 115$ this error is large: in the worst case, heat flowing across a layered cell with $\varepsilon = L/50$, the computed conductivity is $0.063 \text{ W m}^{-1}\text{K}^{-1}$ where the exact value for the same geometry is 0.040 , 59% too high only because the interface has a numerical width. And since the error scales with ε times the interface area per unit volume, it changes as sintering reduces that area, so it distorts the time trend of $\mathbf{K}_{\text{eff}}$, not just its level. We fix it by giving the band a tensor conductivity, arithmetic along the interface and harmonic across it. This is the exact conductivity of a fine ice–air laminate lying parallel to the interface, and it enforces inside the band the same two conditions a sharp interface imposes (temperature continuous along it, heat flux continuous through it), so it cancels both first-order error terms with no fitted parameter, while leaving the bulk phases, the sharp limit and the metamorphism model untouched. Against closed-form sharp solutions it reproduces a laminate’s conductivity to 4×10^{-7} at every interface width tested, where the arithmetic law is 3.8–59% high, and a disk array’s to 0.03% at $\varepsilon = L/800$ with the error falling at second order in ε , while the arithmetic law’s measured error matches our first-order prediction to within 1%.

1 The problem

The effective conductivity of a periodic ice/air cell is obtained from the corrector problem

$$\nabla \cdot (\mathbf{K}(\nabla t_m + \mathbf{e}_m)) = 0 \text{ in } \Omega, \quad t_m \text{ periodic}, \quad \mathbf{K}_{\text{eff}} \mathbf{e}_m = \langle \mathbf{K}(\nabla t_m + \mathbf{e}_m) \rangle, \quad (1)$$

for each direction m . With a sharp interface, $\mathbf{K} = K_{\star} \mathbf{I}$, where $K_{\star} = K_{\text{i}}$ in ice and K_{a} in air; this is the problem of Calonne et al. (2011). With a phase field φ the coefficient must also be defined *inside* the diffuse band, and the usual choice is the arithmetic law $K_{\text{ar}}(\varphi) = K_{\text{a}} + (K_{\text{i}} - K_{\text{a}})\varphi$. The two problems coincide exactly when φ is the indicator function of the ice (companion note, Prop. 1). At finite ε they differ, by an amount of order ε that depends on the interpolation law. For $K_{\text{i}}/K_{\text{a}} = 114.5$ this bias is large: +17% at $\varepsilon = L/128$ for a laminate loaded normal to its layers.

The interpolation is invisible to the sharp problem, since every law with $K(0) = K_{\text{a}}$, $K(1) = K_{\text{i}}$ agrees on the indicator. It is therefore free to choose, and we choose it to cancel the first-order bias.

2 Structure of the diffuse band

Let s be the signed distance to the interface Γ (the $\varphi = \frac{1}{2}$ level set), positive in ice, and let the profile be logistic,

$$\varphi = \ell(s/\varepsilon), \quad \ell(u) = \frac{1}{1 + e^{-u}}, \quad \ell(-u) = 1 - \ell(u). \quad (2)$$

where ℓ is the logistic function. It only describes the shape of the profile, and is unrelated to the excesses Σ_t , Σ_n defined in §3. When ε is small compared with the interface’s radius of curvature

and with the spacing between interfaces, the band is locally a flat one-dimensional layer. Write the temperature gradient $\mathbf{E} = \nabla T$ and the flux $\mathbf{q} = \mathbf{K}\mathbf{E}$. At leading order in the stretched coordinate $u = s/\varepsilon$, two quantities are constant across the layer:

- the *tangential gradient* $\mathbf{E}_t$, because $\mathbf{E}$ is a gradient (curl-free), so its tangential part cannot vary over the short normal distance $O(\varepsilon)$;
- the *normal flux* $q_n = \mathbf{q} \cdot \hat{\mathbf{n}}$, because $\nabla \cdot \mathbf{q} = 0$ and the tangential flux divergence is $O(1)$ while ∂_s is $O(\varepsilon^{-1})$.

These are exactly the quantities that the sharp interface conditions make continuous: the temperature (hence $\mathbf{E}_t$) and the normal flux. Inside the band, then, the tangential sub-layers act in *parallel* and the normal ones in *series*.

3 Surface excesses and the first-order bias

Because the band acts in parallel along Γ and in series across it, it is equivalent, to first order, to a sharp interface carrying two excesses per unit area:

$$\Sigma_t = \int_{-\infty}^{\infty} [K_t(\varphi) - K_\star] ds, \quad \Sigma_n = \int_{-\infty}^{\infty} \left[\frac{1}{K_n(\varphi)} - \frac{1}{K_\star} \right] ds, \quad (3)$$

where K_t and K_n are the band's conductivities along and across the interface ($K_t = K_n = K$ for a scalar law). Σ_t is an excess *conductance*, Σ_n an excess *resistance*. This is the imperfect-interface (thin-interphase) limit of Hashin (2001) and Benveniste (2006).

What the two numbers mean. Take a patch of interface of unit area and follow a line straight through the band. Along the interface the sub-layers conduct side by side (in parallel), so the patch's conductance per unit length is $\int K_t ds$; a sharp interface would give $\int K_\star ds$. Σ_t is the difference: the conductance the diffuse band adds, or removes if negative, compared with a sharp interface. Across the interface the sub-layers are stacked (in series), so what adds is resistance, $\int 1/K_n ds$, and Σ_n is the resistance the band adds compared with a sharp interface. Both integrands vanish outside the band, so both integrals are finite. They are the conduction analogue of Gibbs surface excesses.

Proposition 1. *For a unit mean gradient along $\mathbf{e}$,*

$$\mathbf{e} \cdot \mathbf{K}_{\text{eff}}^\varepsilon \mathbf{e} - \mathbf{e} \cdot \mathbf{K}_{\text{eff}}^\star \mathbf{e} = \frac{1}{|\Omega|} \int_\Gamma \left(\Sigma_t |\mathbf{E}_t|^2 - \Sigma_n q_n^2 \right) d\Gamma + O(\varepsilon^2), \quad (4)$$

with $\mathbf{E}_t$ and q_n taken from the sharp solution.

Justification. $\mathbf{e} \cdot \mathbf{K}_{\text{eff}}^\varepsilon \mathbf{e}$ is a minimum of the energy $\langle \mathbf{E} \cdot \mathbf{K}\mathbf{E} \rangle$ (Dirichlet principle) and, dually, a maximum in terms of the flux (Thomson principle). Split the band's energy density as $K_t E_t^2 + q_n^2/K_n$. A change in tangential conductance at fixed $\mathbf{E}_t$ alters the energy by $\Sigma_t E_t^2$. A change in normal resistance at fixed q_n alters the complementary energy by $\Sigma_n q_n^2$, which changes the conductance by $-\Sigma_n q_n^2$. By stationarity, the accompanying change in the field enters only at second order. The companion note (Eq. 21) gives a rigorous two-sided bound of the same quantity that does not rely on the expansion.

Check (laminate). For layers of period L with n_Γ interfaces, a gradient normal to the layers has $\mathbf{E}_t = 0$ and $q_n = k_\perp$, and (4) gives $\delta \langle K^{-1} \rangle = \Sigma_n n_\Gamma / L$, exactly the closed form. A gradient along the layers has $q_n = 0$, $|\mathbf{E}_t| = 1$, and gives $\delta k_\parallel = \Sigma_t n_\Gamma / L$.

4 Excesses of the scalar laws

This section evaluates the two integrals (3) for a scalar law $K(\varphi)$, in four steps.

Step 1: every excess is proportional to ε . Across the band, the diffuse coefficient is $K(\ell(u))$ and the sharp one is $K(H(u))$, with $u = s/\varepsilon$ and H the unit step ($H = 1$ in ice, 0 in air). This is Lemma 1 of the companion note: the sharp coefficient is the same law evaluated at the indicator. Changing variables $s = \varepsilon u$ in (3) gives

$$\Sigma_t = \varepsilon \int_{-\infty}^{\infty} [K(\ell(u)) - K(H(u))] du, \quad \Sigma_n = \varepsilon \int_{-\infty}^{\infty} \left[\frac{1}{K(\ell(u))} - \frac{1}{K(H(u))} \right] du. \quad (5)$$

The integrals are pure numbers set by K_i/K_a alone. So each excess is ε times a constant, and by (4) the bias in $\mathbf{K}_{\text{eff}}$ is first order in ε unless that constant is zero.

Step 2: a quantity affine in φ has zero excess. Suppose the integrand is $f(\varphi) = \alpha + \beta\varphi$. Then $f(\ell(u)) - f(H(u)) = \beta[\ell(u) - H(u)]$, and $\ell - H$ is an *odd* function. For $u > 0$, $\ell(u) - H(u) = \ell(u) - 1 = -\ell(-u)$ by the antisymmetry in (2), and $\ell(-u) - H(-u) = \ell(-u)$. So the positive lobe on the air side and the negative lobe on the ice side have equal area (Fig. 1), and

$$\int_{-\infty}^{\infty} [\ell(u) - H(u)] du = 0 \implies \text{excess of } f = 0. \quad (6)$$

Physically, a diffuse band that is exactly antisymmetric about $\varphi = \frac{1}{2}$ adds as much ice on the air side as it removes on the ice side.

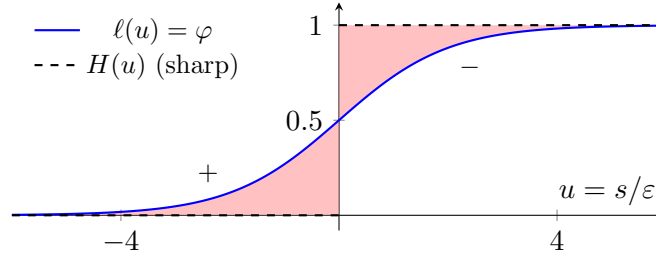


Figure 1: The profile $\ell(u)$ against the sharp step $H(u)$. The two shaded lobes, $\ell - H$ on either side of the interface, are mirror images with opposite sign, so any quantity affine in φ has zero excess.

Apply this to the two laws.

- *Arithmetic*, $K_{\text{ar}} = K_a + (K_i - K_a)\varphi$, is affine, so $\Sigma_t = 0$. But $1/K_{\text{ar}}$ is not affine, so Σ_n must be computed.
- *Harmonic*, $1/K_h = \varphi/K_i + (1 - \varphi)/K_a$, is affine, so $\Sigma_n = 0$. But K_h is not affine, so Σ_t must be computed.

Step 3: computing the nonzero excess. Take Σ_n for the arithmetic law; write $a = K_a$, $b = K_i$, $c = b - a$, so $K(p) = a + cp$. Substitute $p = \ell(u)$. Since $\ell'(u) = \ell(1 - \ell)$, $du = dp/[p(1 - p)]$, and $u \gtrless 0$ becomes $p \gtrless \frac{1}{2}$:

$$\frac{\Sigma_n}{\varepsilon} = \int_0^{1/2} \left[\frac{1}{K(p)} - \frac{1}{a} \right] \frac{dp}{p(1-p)} + \int_{1/2}^1 \left[\frac{1}{K(p)} - \frac{1}{b} \right] \frac{dp}{p(1-p)}. \quad (7)$$

Each bracket vanishes at its own endpoint, which cancels one factor of the denominator: $\frac{1}{K} - \frac{1}{a} = -\frac{cp}{aK}$ and $\frac{1}{K} - \frac{1}{b} = \frac{c(1-p)}{bK}$. So

$$\frac{\Sigma_n}{\varepsilon} = -\frac{c}{a} \int_0^{1/2} \frac{dp}{K(p)(1-p)} + \frac{c}{b} \int_{1/2}^1 \frac{dp}{K(p)p}. \quad (8)$$

Partial fractions, $\frac{1}{K(1-p)} = \frac{1}{b} \left[\frac{c}{K} + \frac{1}{1-p} \right]$ and $\frac{1}{Kp} = \frac{1}{a} \left[\frac{1}{p} - \frac{c}{K} \right]$, give elementary logarithms. With $K_m = K(\frac{1}{2})$,

$$\frac{\Sigma_n}{\varepsilon} = \frac{c}{ab} \left[-\ln \frac{K_m}{a} - \ln 2 + \ln 2 - \ln \frac{b}{K_m} \right] = -\frac{c}{ab} \ln \frac{b}{a}, \quad (9)$$

that is,

$$\Sigma_n^{\text{ar}} = -\varepsilon \left(\frac{1}{K_a} - \frac{1}{K_i} \right) \ln \frac{K_i}{K_a}. \quad (10)$$

The harmonic law's Σ_t is the same calculation with the roles of K and $1/K$ exchanged (conductivity $\leftrightarrow$ resistivity), which gives

$$\Sigma_t^{\text{h}} = -\varepsilon (K_i - K_a) \ln \frac{K_i}{K_a}. \quad (11)$$

Both were checked against direct quadrature of (5).

Step 4: summary and meaning.

law	Σ_t (along)	Σ_n (across)
arithmetic	0	$-\varepsilon (K_a^{-1} - K_i^{-1}) \ln(K_i/K_a)$
harmonic	$-\varepsilon (K_i - K_a) \ln(K_i/K_a)$	0

Both nonzero entries are negative. For the arithmetic law that is a *missing resistance* across the interface. Put into (4) with its minus sign, it makes the cell conduct too well across interfaces. For the harmonic law it is a *missing conductance* along the interface, so the cell conducts too poorly along them. At $K_i/K_a = 114.5$ the constants are $234.96 \varepsilon \text{ m K W}^{-1}$ and $10.76 \varepsilon \text{ W m}^{-1} \text{ K}^{-1}$.

No scalar law can make both entries zero: by Step 2 that would need K and $1/K$ to be affine in φ at the same time, and only a constant is. A direction-dependent (tensor) coefficient can, since it only needs K affine along the interface and $1/K$ affine across it. That is the next section.

5 The tensor law

Use each law in the direction where its excess vanishes:

$\mathbf{K}(\varphi, \nabla \varphi) = K_{\text{ar}}(\varphi) (\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_{\text{h}}(\varphi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \frac{\nabla \varphi}{|\nabla \varphi|}.$

(12)

Then $K_t = K_{\text{ar}}$ and $K_n = K_{\text{h}}$ in (3), so

$$\Sigma_t = \Sigma_n = 0, \quad \mathbf{K}_{\text{eff}}^\varepsilon - \mathbf{K}_{\text{eff}}^* = O(\varepsilon^2). \quad (13)$$

Interpretation: a microscopic laminate. (12) is not an ad hoc blend. It is the exact effective conductivity of a finely layered mixture of ice and air, with ice fraction φ , whose layers lie parallel to the interface (layer normal $\hat{\mathbf{n}}$). For such a laminate, conduction along the layers is the arithmetic mean and conduction across them is the harmonic mean (Wiener's bounds; Milton 2002, ch. 9). The tensor law therefore models each point of the band as a laminate oriented with the interface, which is what the band locally is.

Properties.

1. *Same sharp problem.* At $\varphi \in \{0, 1\}$ both branches equal K_* , so $\mathbf{K} = K_* \mathbf{I}$ in the bulk and the sharp limit is untouched (companion note, Lemma 1). Only the band changes.
2. *Well posed.* $\mathbf{K}$ is symmetric with eigenvalues K_{ar} (tangential) and K_{h} (normal), both in $[K_{\text{a}}, K_{\text{i}}]$. The cell problem remains uniformly elliptic with the same bounds, and its discrete operator remains symmetric positive definite.
3. *No free parameters.* Nothing is fitted or calibrated. The law follows from the two excess conditions alone and holds for any $K_{\text{i}}/K_{\text{a}}$.
4. *Undefined normal is harmless.* $\hat{\mathbf{n}}$ is undefined where $\nabla\varphi = 0$, which happens in the bulk, where $K_{\text{ar}} = K_{\text{h}}$ and $\mathbf{K}$ is isotropic whatever $\hat{\mathbf{n}}$ is. We set $\mathbf{K} = K_{\text{ar}} \mathbf{I}$ there.

6 Why this is legitimate

The corrector problem (1) is post-processing of a *frozen* microstructure. The phase-field evolution (vapour transport, latent heat, interface kinetics) is solved with its own interpolation and is unaffected. The two interpolations answer different questions. The evolution law sets how the geometry develops; the homogenization law sets how a given geometry is turned into $\mathbf{K}_{\text{eff}}$. Only the second is changed. The same laws already give the same sharp problem, so the change amounts to choosing, among the band representations of that problem, the one with no first-order error.

This parallels the thin-interface corrections of Karma & Rappel (1998), Almgren (1999) and Kaempfer & Plapp (2009), which remove $O(\varepsilon)$ artifacts from phase-field *dynamics* (for example, with an anti-trapping current or a renormalized kinetic coefficient). Here the problem is static and has no source term, so the thin-interface analysis stops at the two equilibrium excesses (3), and it closes in elementary form. The comparison is one of method only. The tensor law is a property of the steady-state homogenization, and it is not an argument for changing the energy balance of any metamorphism model.

7 Verification

Two cells with sharp closed forms, run over an ε ladder at fixed $\varepsilon/h = 5.12$ (so the interface bias is not confounded with mesh convergence):

1. *Laminate* ($\bar{\varphi} = \frac{1}{2}$, $n_{\Gamma} = 2$). Along the layers $k_{\parallel} = \langle K_{\text{ar}} \rangle$ is exact for both laws. Across the layers $1/k_{\perp} = \langle 1/K_n \rangle$ is linear in ε with slope $(n_{\Gamma}/L) |\Sigma_n|/\varepsilon$ for the arithmetic law, and exactly flat at the sharp value for the tensor law. The laminate's normal is a fixed coordinate axis, so this case does not exercise the off-diagonal part of (12).
2. *Disk array* (square array, $f = \pi R^2/L^2 = 0.196$). The sharp value is Rayleigh's, $k = 0.0295684 \text{ W m}^{-1} \text{ K}^{-1}$ (Perrins et al. 1979); Maxwell-Garnett agrees to 1.8×10^{-4} . Here the normal takes every direction. Using the dipole field in (4) gives the arithmetic law's slope with no fitted constants,

$$\frac{dk}{d\varepsilon} = \frac{|\Sigma_n|}{\varepsilon} \frac{\pi R}{L^2} \left[\frac{K_{\text{a}}(1 + \beta)}{1 - \beta f} \right]^2, \quad \beta = \frac{K_{\text{i}} - K_{\text{a}}}{K_{\text{i}} + K_{\text{a}}}, \quad (14)$$

which is +15% at $\varepsilon = L/100$ and +1.9% at $L/800$. The disk ladder starts at $L/100$: at $L/50$, $\varepsilon/R = 0.08$ and the visible band spans most of the radius, outside the thin-interface regime (measured there: arithmetic +46%, tensor +6.7%).

ε	laminate, $k_{\perp}$ bias	
	arithmetic (pred.)	tensor (meas.)
$L/50$	+59.4%	$< 4 \times 10^{-7}$
$L/64$	+41.1%	$< 4 \times 10^{-7}$
$L/128$	+17.0%	$< 4 \times 10^{-7}$
$L/256$	+7.9%	$< 4 \times 10^{-7}$
$L/512$	+3.8%	$< 4 \times 10^{-7}$

On the laminate the tensor law is exact to solver precision at every rung, as predicted. The fitted slope of $1/k_{\perp}$ is 0.10% of the arithmetic law's.

ε	disk array, k bias			
	arith. (pred.)	arith. (meas.)	tensor (meas.)	tensor, IC offset removed
$L/100$	+15.07%	+18.46%	+1.377%	+1.219%
$L/200$	+7.53%	+8.37%	+0.283%	+0.205%
$L/400$	+3.77%	+3.99%	+0.073%	+0.034%
$L/800$	+1.88%	+1.95%	+0.026%	+0.006%

For the arithmetic law, a quadratic fit in ε gives a first-order coefficient of $440.9 \text{ W m}^{-2} \text{K}^{-1}$ against the predicted 445.5 (−1.0%). The intercept is within 0.04% of Rayleigh's value. This confirms Prop. 1 on a curved interface with no fitted constants. For the tensor law the first-order coefficient is −2.5% of the arithmetic one, and at every rung the error is at most $0.075\times$ the arithmetic error. The last column removes an offset in the benchmark's initial condition, which stretched the disk by $(N+1)/N$ on an N^2 mesh (measured area excess $2/N$ to four digits). It has since been fixed in the code, so a rerun measures this column directly. With the offset removed, the tensor residual falls by about $6\times$ per halving of ε (observed order ≈ 2.5), consistent with (13).

Drivers: `enceladus_DSM/studies/keff_sharp_limit/verification/` (`--interp tensor`) and `.../disk/`.

8 Scope and limitations

- *Residual* $O(\varepsilon^2)$. This includes curvature terms and a geometric one: a diffuse disk holds $\pi^3\varepsilon^2/3$ more area of ice than the sharp disk.
- *Thin ligaments*. The derivation assumes the band is a flat 1D layer, which fails where two interfaces are closer than the band width (about $6-9\varepsilon$), as in young sinter necks. No interpolation law is justified there. This limits how well the phase field resolves the geometry, and it applies to the evolution model's geometry as much as to $\mathbf{K}_{\text{eff}}$.
- *Profile*. The zero excesses use the antisymmetry of the logistic profile. Any profile with $\varphi(-s) = 1 - \varphi(s)$ about the $\varphi = \frac{1}{2}$ level set gives the same result; a profile distorted by the evolution changes it only at the next order.
- *Geometry error is untouched*. The law removes the bias of homogenizing a given φ . It does not correct errors in φ itself.

9 Related work

The ingredients are established; their combination for this cell problem, with the closed-form excesses, is what this note adds.

- *Laminates and mixing bounds.* Arithmetic along and harmonic across layers is Wiener’s classical result. Milton (2002) and Torquato (2002) are the standard texts on homogenization, laminates and bounds on $\mathbf{K}_{\text{eff}}$.
- *Thin interphases as imperfect interfaces.* A thin layer replaced by a sharp interface with jump conditions, to first order in its thickness, is developed for conduction by Hashin (2001) and in general form by Benveniste (2006). Eq. (4) is the effective-property consequence of that replacement.
- *Excess quantities in thin-interface phase-field analysis.* Karma & Rappel (1998), Almgren (1999), McFadden, Wheeler & Anderson (2000) and Elder et al. (2001) show that $O(\varepsilon)$ errors of diffuse-interface models are integrals of the form (3), and that interpolation functions can be chosen to make them vanish. McFadden et al. treat unequal conductivities in particular.
- *Tensor (rank-one) interpolation in phase-field models.* The closest analogue is in phase-field elasticity. Durga, Wollants & Moelans (2013) show that Voigt and Reuss interpolation of stiffness create interfacial excess energies. Mosler, Shchyglo & Montazer Hojjat (2014) and Schneider et al. (2015) remove them by interpolating differently along and across $\hat{\mathbf{n}}$, so the mechanical jump conditions hold inside the band. That is continuous normal traction and tangential strain, the elastic counterparts of continuous q_n and $\mathbf{E}_t$ here.
- *Diffuse-domain methods.* Li et al. (2009) and Lervåg & Lowengrub (2015) analyse elliptic problems posed with a smoothed indicator and show, by matched asymptotics, which formulations are first- or second-order accurate in ε .

10 Suggested Methods text

The effective conductivity tensor is obtained by periodic homogenization (Calonne et al., 2011), solving the corrector problem on the frozen phase field at each sampled time. Inside the diffuse interface the conductivity must be interpolated. A scalar interpolation carries a first-order bias in the interface width ε : to first order the band behaves as a sharp interface with an excess tangential conductance Σ_t and an excess normal resistance Σ_n (Hashin, 2001; Benveniste, 2006). For the logistic profile, the arithmetic law gives $\Sigma_t = 0$ but $\Sigma_n = -\varepsilon(K_a^{-1} - K_i^{-1}) \ln(K_i/K_a)$, which overestimates conduction across interfaces by up to tens of percent at $K_i/K_a \approx 115$. The harmonic law has the opposite defect. We therefore use the tensor law $\mathbf{K} = K_{\text{ar}}(\varphi)(\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_{\text{h}}(\varphi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}$, $\hat{\mathbf{n}} = \nabla\varphi/|\nabla\varphi|$, whose excesses both vanish, so the error is $O(\varepsilon^2)$. It reduces to the sharp conductivities in each bulk phase, has no adjustable parameters, and affects only the homogenization step, not the phase-field evolution. We verified it against closed-form sharp solutions for a laminate and for a square array of disks (Rayleigh, 1892; Perrins et al., 1979).

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